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17.10 Tables and figures

Table 17.1. Sole in Division 7.d. Official landings (tonnes) by country over the period 1974–2024; ICES estimates (as reported in InterCatch) for both landings and discards (tonnes) used by the working group. TAC (tonnes) represents landings until 2015. From 2016 onwards, TAC represents catch. * including BMS.

Year	Official Landings					ICES estimates		TAC
	Belgium	France	UK (E&W)	Other	Total	Landings	Discards	
1974	159	383	309	3	854	884		
1975	132	464	244	1	841	882		
1976	203	599	404		1206	1305		
1977	225	737	315		1277	1335		
1978	241	782	366		1389	1589		
1979	311	1129	402		1842	2215		
1980	302	1075	159		1536	1923		
1981	464	1513	160		2137	2477		
1982	525	1828	317	4	2674	3190	196	
1983	502	1120	419		2041	3458	101	
1984	592	1309	505		2406	3575	141	
1985	568	2545	520		3633	3837	242	
1986	858	1528	551		2937	3932	145	
1987	1100	2086	655		3841	4791	197	3850

Year	Official Landings				ICES estimates		TAC	
	Belgium	France	UK (E&W)	Other	Total	Landings		Discards
1988	667	2057	578		3302	3853	198	3850
1989	646	1610	689		2945	3805	192	3850
1990	996	1255	785		3036	3647	342	3850
1991	904	2054	826		3784	4351	368	3850
1992	891	2187	706	10	3794	4072	275	3500
1993	917	2322	610	13	3862	4299	273	3200
1994	940	2382	701	15	4038	4383	122	3800
1995	817	2248	669	9	3743	4420	282	3800
1996	899	2322	877		4098	4797	174	3500
1997	1306	1702	933		3941	4764	147	5230
1998	541	1703	803		3047	3363	127	5230
1999	880	2251	769		3900	4135	247	4700
2000	1021	2190	621		3832	3476	201	4100
2001	1313	2482	822		4617	4025	317	4600
2002	1643	2780	976		5399	4733	444	5200
2003	1657	3475	1114	1	6247	6977	584	5400
2004	1485	3070	1112		5667	5819	258	5900
2005	1221	2832	567		4620	4748	344	5700
2006	1547	2627	658	0	4832	4830	315	5720
2007	1530	2981	801	1	5313	5421	332	6220
2008	1368	2880	724	0	4972	4963	183	6590
2009	1475	3047	760	0	5282	4828	287	5274
2010	1294	2476	679	0	4449	4108	273	4219
2011	1222	2281	700	0	4203	4136	342	4852
2012	941	2475	627	0	4043	4058	445	5580
2013	952	2884	605	0	4441	4295	180	5900
2014	1496	2507	648	< 1	4651	4626	216	4838
2015	1048	1895	468	0	3411	3385	263	3483

Year	Official Landings				ICES estimates			TAC
	Belgium	France	UK (E&W)	Other	Total	Landings	Discards	
2016	799	1337	392	< 1	2528	2433	106	3258
2017	697*	1178	349	< 1	2224	2090	156	2724
2018	653	1265	394	< 1	2312	2395	263	3405
2019	603*	925	245*	< 1	1773	1648	404	2515
2020	686*	827	199*	< 1	1712	1562	409*	2797
2021	623*	824	233*	1	1681	1567	348*	3248
2022	644*	884*	262	1	1791	1699	316	2380
2023	538*	567	184	< 1	1289	1293	211	1747
2024	412*	519*	136	< 1	1067	1112	471	1504
2025								1209

Table 17.2. Sole in Division 7.d. Summary of the InterCatch data in 2024 (imported vs. raised data; sampled vs. estimated data).

CatchCategory	RaisedOrImported	SampledOrEstimated	CATON	perc
Landings	Imported_Data	Sampled_Distribution	953.2	86
Landings	Imported_Data	Estimated_Distribution	158.6	14
Discards	Imported_Data	Sampled_Distribution	343.8	73
Discards	Imported_Data	Estimated_Distribution	0.298	0
Discards	Raised_Discards	Estimated_Distribution	127	27
BMS landing	Imported_Data	Estimated_Distribution	0	NA

Table 17.3. Sole in Division 7.d. Landings percentages by gear type for 2017–2024 (GNS/GTR = gill and trammel nets; TBB = beam trawls; OTB/OTT/SSC/SDN = otter trawls and seines; other gears include MIS/FPO/DRB/LHM/LLS).

Landings by gear	2017	2018	2019	2020	2021	2022	2023	2024
GNS/GTR	46%	44%	33%	28%	32%	31%	29%	30%
TBB	31%	28%	33%	36%	35%	34%	45%	43%
OTB/OTT/SSC/SDN	17.6%	24%	30%	33%	30%	30%	24%	21%
Other	4.7%	4.5%	3.7%	2.8%	2.7%	3.9%	2.6%	7%

Table 17.4. Sole in Division 7.d. Catch numbers at age (in thousands).

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
1	375	0	59	122	122	22	236	405	3092	952	261	211	77	2082
2	3432	1136	2630	4961	1685	4197	2910	4686	3836	9646	5446	6769	934	4006
3	5688	3812	3476	5795	5904	4158	7995	3586	6214	4575	9794	7179	6912	4874
4	1710	3971	2630	1675	3259	3336	1633	4482	1172	4242	1925	5551	6017	4857
5	558	895	1890	1032	911	2068	1167	1443	1505	608	2006	1015	3427	2987
6	636	731	736	1863	771	1046	859	842	302	1000	289	565	586	1986
7	535	624	454	145	1062	1095	390	574	392	258	370	163	570	377
8	233	330	313	158	155	785	255	201	260	247	135	188	109	278
9	118	107	134	156	190	111	256	166	129	258	171	116	147	88
10	81	88	98	69	212	163	83	224	126	92	95	62	93	106
11+	196	191	235	128	372	459	275	282	489	382	231	129	258	241

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
1	22	60	82	417	343	418	1756	57	121	771	412	168	826	1270
2	2456	2004	1855	4395	4831	8136	9431	15482	4164	6957	6942	7620	2872	4446
3	8650	6761	6259	9470	5412	6905	8367	10669	11013	5185	4285	8307	8562	4494
4	3094	5106	2761	3369	4485	1627	3839	4069	3682	4777	3097	3169	5679	6164
5	3227	2096	1649	1319	1084	2509	1422	2168	4595	1256	3316	1794	1452	2500
6	1830	1676	612	871	507	731	657	656	1670	920	1207	2769	1086	808
7	1289	920	562	352	320	291	299	2068	379	636	1128	1010	758	719
8	271	776	443	672	148	128	129	229	394	392	579	753	410	664
9	319	239	354	351	328	56	97	73	291	211	239	450	157	277
10	112	169	239	192	150	81	57	134	254	104	233	194	168	239
11+	344	267	301	359	248	265	197	285	443	266	383	473	276	425

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	585	353	739	40	372	300	144	565	1643	664	7	1040	433	545	1176
2	5827	6148	3759	1150	1244	2131	1145	1060	3378	2341	2736	1119	2164	1314	3431
3	4255	6938	8544	5951	3502	2101	2185	2467	1846	3086	2683	3426	1490	2020	1419

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
4	2953	2854	5253	6595	6639	2303	1253	1447	2626	1086	2223	1705	2512	801	1133
5	3034	1562	1433	2539	4259	3496	1308	826	1022	1476	812	1225	1461	1375	431
6	1621	1469	930	762	1853	2555	1553	876	736	481	915	471	844	447	650
7	320	562	563	545	687	1194	1059	850	619	312	427	396	441	400	239
8	277	178	414	535	417	463	598	698	821	227	166	244	385	105	183
9	288	147	98	205	374	142	188	287	451	392	131	110	197	163	49
10	102	132	46	59	145	238	211	139	286	282	182	75	74	74	79
11+	376	179	259	129	255	272	322	194	292	230	331	224	221	118	115

Table 17.5. Sole in Division 7.d. Catch weights at age (kg)

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
1	0.078	0.000	0.076	0.068	0.103	0.073	0.078	0.081	0.091	0.087	0.078	0.065	0.076	0.099
2	0.155	0.157	0.162	0.166	0.164	0.159	0.138	0.140	0.162	0.146	0.139	0.134	0.136	0.160
3	0.215	0.220	0.224	0.220	0.203	0.226	0.216	0.184	0.228	0.199	0.194	0.189	0.178	0.171
4	0.309	0.299	0.311	0.279	0.303	0.292	0.276	0.269	0.287	0.264	0.265	0.245	0.233	0.228
5	0.385	0.403	0.379	0.367	0.362	0.352	0.359	0.292	0.348	0.353	0.289	0.334	0.287	0.254
6	0.427	0.435	0.435	0.393	0.386	0.406	0.408	0.357	0.339	0.393	0.402	0.383	0.354	0.332
7	0.439	0.434	0.416	0.515	0.436	0.410	0.458	0.387	0.469	0.421	0.390	0.536	0.380	0.356
8	0.509	0.524	0.538	0.543	0.520	0.482	0.514	0.472	0.465	0.430	0.462	0.553	0.505	0.385
9	0.502	0.537	0.529	0.594	0.502	0.465	0.553	0.515	0.487	0.434	0.459	0.515	0.484	0.490
10	0.463	0.583	0.565	0.595	0.523	0.538	0.563	0.547	0.518	0.478	0.463	0.766	0.496	0.494
11+	0.672	0.628	0.714	0.800	0.602	0.618	0.665	0.701	0.562	0.566	0.566	0.667	0.616	0.654

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
1	0.109	0.106	0.101	0.099	0.110	0.082	0.091	0.101	0.097	0.123	0.133	0.095	0.121	0.113
2	0.150	0.139	0.145	0.137	0.129	0.138	0.147	0.148	0.147	0.158	0.150	0.158	0.156	0.155
3	0.170	0.180	0.165	0.181	0.169	0.202	0.195	0.218	0.181	0.191	0.192	0.175	0.207	0.201
4	0.227	0.231	0.233	0.213	0.221	0.281	0.251	0.286	0.238	0.262	0.233	0.201	0.243	0.252
5	0.268	0.291	0.285	0.259	0.331	0.287	0.315	0.365	0.269	0.353	0.286	0.267	0.258	0.268
6	0.323	0.342	0.343	0.280	0.376	0.333	0.375	0.407	0.293	0.434	0.338	0.280	0.311	0.322

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
7	0.360	0.389	0.382	0.290	0.424	0.367	0.376	0.165	0.410	0.455	0.394	0.339	0.370	0.316
8	0.405	0.404	0.417	0.341	0.427	0.374	0.393	0.474	0.449	0.490	0.425	0.387	0.397	0.383
9	0.435	0.503	0.484	0.358	0.384	0.493	0.469	0.424	0.390	0.566	0.562	0.452	0.433	0.383
10	0.465	0.474	0.435	0.374	0.459	0.511	0.420	0.504	0.487	0.648	0.497	0.424	0.511	0.430
11+	0.585	0.651	0.616	0.535	0.680	0.544	0.531	0.565	0.664	0.550	0.552	0.570	0.509	0.484

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	0.101	0.089	0.058	0.095	0.098	0.077	0.093	0.094	0.081	0.058	0.091	0.083	0.094	0.133	0.099
2	0.152	0.148	0.121	0.141	0.134	0.127	0.136	0.143	0.121	0.118	0.114	0.124	0.132	0.129	0.131
3	0.192	0.201	0.181	0.189	0.176	0.168	0.199	0.184	0.171	0.175	0.152	0.164	0.153	0.174	0.171
4	0.241	0.245	0.233	0.241	0.231	0.223	0.242	0.229	0.214	0.230	0.197	0.195	0.193	0.217	0.225
5	0.276	0.301	0.270	0.297	0.267	0.266	0.266	0.252	0.268	0.244	0.224	0.240	0.233	0.223	0.253
6	0.322	0.330	0.312	0.301	0.325	0.282	0.285	0.291	0.289	0.274	0.243	0.277	0.265	0.280	0.272
7	0.334	0.357	0.375	0.384	0.328	0.330	0.320	0.293	0.289	0.290	0.271	0.291	0.255	0.298	0.293
8	0.337	0.424	0.354	0.402	0.389	0.329	0.371	0.362	0.250	0.318	0.312	0.320	0.307	0.356	0.325
9	0.367	0.389	0.424	0.415	0.413	0.408	0.361	0.432	0.327	0.272	0.382	0.373	0.332	0.353	0.397
10	0.520	0.425	0.544	0.463	0.494	0.372	0.358	0.479	0.362	0.338	0.285	0.409	0.372	0.382	0.370
11+	0.502	0.534	0.521	0.572	0.527	0.480	0.436	0.525	0.409	0.394	0.417	0.465	0.402	0.491	0.415

Table 17.6. Sole in Division 7.d. Stock weights at age (kg)

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
1	0.077	0.078	0.075	0.067	0.102	0.072	0.077	0.080	0.090	0.086	0.077	0.064	0.075	0.098	0.108
2	0.167	0.169	0.175	0.179	0.177	0.172	0.149	0.151	0.175	0.157	0.150	0.145	0.147	0.173	0.162
3	0.243	0.248	0.253	0.248	0.229	0.255	0.244	0.208	0.257	0.225	0.219	0.213	0.201	0.193	0.192
4	0.352	0.340	0.354	0.318	0.345	0.332	0.314	0.306	0.327	0.301	0.302	0.279	0.265	0.260	0.258
5	0.424	0.444	0.417	0.404	0.399	0.388	0.395	0.322	0.383	0.389	0.318	0.368	0.316	0.280	0.295
6	0.472	0.481	0.481	0.435	0.427	0.449	0.451	0.395	0.375	0.435	0.445	0.424	0.392	0.367	0.357
7	0.488	0.482	0.462	0.572	0.485	0.456	0.509	0.430	0.521	0.468	0.434	0.596	0.422	0.396	0.400
8	0.441	0.454	0.466	0.471	0.451	0.418	0.445	0.409	0.403	0.373	0.400	0.479	0.438	0.334	0.351
9	0.486	0.520	0.512	0.575	0.486	0.450	0.536	0.499	0.472	0.420	0.445	0.499	0.469	0.475	0.421

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
10	0.481	0.606	0.587	0.618	0.543	0.559	0.585	0.568	0.538	0.497	0.481	0.796	0.515	0.513	0.483
11+	0.720	0.676	0.766	0.861	0.646	0.662	0.713	0.754	0.604	0.607	0.606	0.718	0.662	0.701	0.629

age	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
1	0.105	0.100	0.098	0.109	0.081	0.090	0.100	0.097	0.123	0.136	0.087	0.115	0.113	0.101	0.089
2	0.150	0.156	0.148	0.139	0.149	0.159	0.160	0.142	0.154	0.142	0.140	0.141	0.148	0.129	0.119
3	0.203	0.186	0.204	0.191	0.228	0.220	0.246	0.179	0.189	0.182	0.168	0.190	0.196	0.193	0.188
4	0.263	0.265	0.242	0.252	0.320	0.286	0.326	0.225	0.262	0.241	0.220	0.246	0.264	0.260	0.250
5	0.320	0.314	0.285	0.365	0.316	0.347	0.402	0.265	0.361	0.316	0.282	0.249	0.288	0.292	0.313
6	0.378	0.379	0.310	0.416	0.368	0.415	0.450	0.285	0.443	0.352	0.315	0.338	0.344	0.363	0.338
7	0.432	0.425	0.322	0.471	0.408	0.418	0.183	0.409	0.466	0.398	0.346	0.333	0.304	0.363	0.371
8	0.350	0.361	0.295	0.370	0.324	0.341	0.411	0.510	0.514	0.465	0.443	0.435	0.438	0.371	0.481
9	0.487	0.469	0.347	0.372	0.477	0.454	0.411	0.391	0.598	0.574	0.496	0.373	0.352	0.421	0.409
10	0.493	0.452	0.389	0.477	0.531	0.436	0.524	0.514	0.704	0.496	0.397	0.713	0.437	0.542	0.458
11+	0.700	0.661	0.577	0.727	0.585	0.569	0.608	0.692	0.588	0.632	0.613	0.472	0.606	0.537	0.561

Age	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	0.057	0.092	0.097	0.075	0.090	0.094	0.081	0.060	0.093	0.082	0.076	0.098	0.098
2	0.104	0.104	0.093	0.102	0.101	0.135	0.111	0.104	0.097	0.101	0.091	0.106	0.125
3	0.177	0.163	0.144	0.177	0.211	0.176	0.177	0.177	0.142	0.173	0.129	0.166	0.167
4	0.228	0.244	0.235	0.244	0.273	0.229	0.247	0.232	0.203	0.220	0.194	0.225	0.224
5	0.275	0.339	0.283	0.296	0.294	0.267	0.296	0.272	0.263	0.274	0.233	0.237	0.257
6	0.331	0.340	0.346	0.308	0.331	0.305	0.324	0.305	0.273	0.311	0.268	0.301	0.306
7	0.387	0.439	0.396	0.373	0.367	0.323	0.343	0.307	0.317	0.331	0.272	0.329	0.374
8	0.384	0.416	0.429	0.336	0.448	0.384	0.332	0.352	0.369	0.371	0.306	0.391	0.368
9	0.467	0.431	0.442	0.398	0.537	0.478	0.371	0.286	0.459	0.398	0.347	0.382	0.502
10	0.548	0.416	0.592	0.380	0.456	0.508	0.407	0.361	0.346	0.416	0.352	0.374	0.470
11+	0.573	0.683	0.541	0.519	0.580	0.575	0.463	0.457	0.495	0.537	0.454	0.557	0.489

Table 17.7. Sole in Division 7.d. Tuning series 1: revised Belgian commercial beam trawl LPUE (2004–2024).

	Effort	Biomass index
2004	1	14.72204251
2005	1	14.95745776
2006	1	16.02364983
2007	1	16.71950689
2008	1	15.28191325
2009	1	19.06356295
2010	1	19.41034228
2011	1	17.32139264
2012	1	18.43170695
2013	1	18.52714202
2014	1	24.89342401
2015	1	18.19680586
2016	1	14.45257677
2017	1	12.01916843
2018	1	12.78568645
2019	1	11.70561923
2020	1	13.37807871
2021	1	12.89287361
2022	1	12.93832509
2023	1	12.13986061
2024	1	11.19071511

Table 17.8. Sole in Division 7.d. Tuning series 2: revised French commercial otter trawl LPUE (2005–2024).

	Effort	Biomass index
2005	1	13.27654273
2006	1	18.98994703
2007	1	16.07989295
2008	1	19.60347081
2009	1	17.86384112

	Effort	Biomass index
2010	1	22.64104298
2011	1	22.29687739
2012	1	18.34728866
2013	1	20.42425728
2014	1	22.77504759
2015	1	15.1899479
2016	1	15.85655742
2017	1	13.68850757
2018	1	19.10079272
2019	1	15.27052429
2020	1	13.89853034
2021	1	13.61437138
2022	1	12.34374796
2023	1	8.166289604
2024	1	5.259793771

Table 17.9. Sole in Division 7.d. Tuning series 3: UK (E&W) commercial beam trawl LPUE (1986–2024).

	Effort	Biomass index
1986	1	138.72492
1987	1	142.8913114
1988	1	133.2336623
1989	1	107.7585952
1990	1	109.8050754
1991	1	71.17088069
1992	1	65.53329156
1993	1	53.93743589
1994	1	55.80003059
1995	1	65.99439722
1996	1	91.56366808

	Effort	Biomass index
1997	1	88.65710825
1998	1	101.4844797
1999	1	94.33305591
2000	1	93.5978889
2001	1	98.05895762
2002	1	128.4165414
2003	1	118.3101589
2004	1	125.8325338
2005	1	141.4056712
2006	1	125.3070046
2007	1	120.9682142
2008	1	108.3932023
2009	1	85.26167953
2010	1	86.47515299
2011	1	82.65608545
2012	1	77.33684063
2013	1	81.5939132
2014	1	87.09997481
2015	1	92.38899153
2016	1	97.06890425
2017	1	76.36326451
2018	1	77.31401802
2019	1	73.77754664
2020	1	86.43207608
2021	1	94.30956124
2022	1	106.8612657
2023	1	100.5415222
2024	1	84.16520822

Table 17.10. Sole in Division 7.d. Tuning series 4: UK (E&W) beam trawl survey (Q3) (1989–2024); grey values are currently not included in the assessment.

	Effort	Age1	Age2	Age3	Age4	Age5	Age6
1989	1	3.01	22.09	4.62	2.45	0.56	0.35
1990	1	17.96	5.55	5.55	1.24	1.01	0.33
1991	1	12.14	31.17	3.19	2.82	0.48	0.67
1992	1	1.33	15.29	13.47	1.07	1.61	0.34
1993	1	0.82	22.96	11.42	9.97	1.14	1.52
1994	1	8.33	4.26	11.07	4.65	4.3	0.28
1995	1	5.89	16.09	2.22	3.51	1.67	2.12
1996	1	5.3	10.79	5.97	1.07	1.86	1.15
1997	1	24.75	10.85	4.42	1.94	0.26	0.82
1998	1	3.27	24.11	3.67	1.47	0.83	0.19
1999	1	35.99	8.22	11.33	1.59	0.73	1.02
2000	1	14.98	27.45	5.52	4.85	1.48	0.68
2001	1	10.19	27.88	11.55	1.67	2.33	0.75
2002	1	53.56	16.11	8.6	5.11	0.45	1.04
2003	1	11.03	45.65	5.87	3.2	2.05	0.42
2004	1	12.67	11.81	10.97	2.08	2.02	1.34
2005	1	43.27	6.91	3.5	5.18	1.9	1.15
2006	1	10.84	42.62	4.51	2.68	2.59	0.55
2007	1	2.57	28.97	15.45	1.47	1.04	1.56
2008	1	3.77	7.35	9.14	5.82	0.4	0.68
2009	1	51.25	19.16	7.1	5.81	5.02	0.44
2010	1	16.59	30.76	5.14	1.66	2.7	2.73
2011	1	13.66	28.6	14.7	1.66	0.54	2.62
2012	1	1.75	9.72	7.51	3.53	0.92	0.39
2013	1	0.72	8.91	15.09	9.72	3.23	1.12
2014	1	25.39	16.35	12.38	11.92	5.09	2.73
2015	1	25.24	21.36	6.04	2.29	4.51	2.08
2016	1	10.17	33.14	11.17	3.16	3.17	3.02

	Effort	Age1	Age2	Age3	Age4	Age5	Age6
2017	1	27.85	15.18	16.26	2.67	2.13	1.52
2018	1	14.86	36.49	6.66	10.32	1.74	2.13
2019	1	56.54	31.08	19.53	1.18	4.01	2.53
2020	1	1.87	42.73	8.01	4.62	1.15	1.84
2021	1	32.95	16.06	22.82	5.31	4.65	1.10
2022	/	/	/	/	/	/	/
2023	1	19.55	15.49	10.26	2.91	6.26	0.84
2024	1	5.52	10.54	4.00	2.36	1.06	2.48

Table 17.11. Sole in Division 7.d. Tuning series 5: UK (E&W) young fish survey (1987–2006).

	Effort	Age1
1987	1	1.38
1988	1	1.87
1989	1	0.62
1990	1	1.9
1991	1	3.69
1992	1	1.5
1993	1	1.33
1994	1	2.68
1995	1	2.91
1996	1	0.57
1997	1	1.12
1998	1	1.12
1999	1	1.47
2000	1	2.47
2001	1	0.38
2002	1	4.15
2003	1	1.44
2004	1	2.72
2005	1	4.07

	Effort	Age1
2006	1	2.21

Table 17.12. Sole in Division 7.d. Tuning series 6: French young fish survey (1987–2024) funded by EDF (noursom).

	Effort	Age1
1987	1	0.07
1988	1	0.17
1989	1	0.14
1990	1	0.54
1991	1	0.38
1992	1	0.22
1993	1	0.03
1994	1	0.7
1995	1	0.28
1996	1	0.15
1997	1	0.03
1998	1	0.1
1999	1	0.35
2000	1	0.31
2001	1	1.21
2002	1	0.11
2003	1	0.32
2004	1	0.15
2005	1	0.82
2006	1	0.83
2007	1	0.08
2008	1	0.06
2009	1	2.78
2010	1	0.1
2011	1	0.32

	Effort	Age1
2012	1	0.35
2013	1	0.052
2014	1	0.04
2015	1	0.09
2016	1	0.04
2017	1	0.05
2018	1	0.03
2019	1	0.45
2020	1	0.38
2021	1	0.07
2022	1	0.01
2023	1	0.14
2024	1	No data point available

Table 17.13. Sole in Division 7.d. SAM model configuration of the 2025 assessment

Where a matrix is specified rows corresponds to fleets and columns to ages.

Same number indicates same parameter used

Numbers (integers) starts from zero and must be consecutive

\$minAge

The minimum age class in the assessment

1

\$maxAge

The maximum age class in the assessment

11

\$maxAgePlusGroup

Is last age group considered a plus group (1 yes, or 0 no).

1 0 0 0 0 0

\$keyLogFsta

Coupling of the fishing mortality states (normally only first row is used).

[,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]

[1,] 0 1 2 3 4 5 5 6 6 7 7

[2,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

[3,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

[4,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

[5,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

[6,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

[7,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

\$corFlag

Correlation of fishing mortality across ages (0 independent, 1 compound symmetry, or 2 AR(1))

0

\$keyLogFpar

Coupling of the survey catchability parameters (normally first row is not used, as that is covered by fishing mortality).

[,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]

```

[1,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[2,] 0 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[3,] 1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[4,] 2 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[5,] 3 4 5 6 7 7 -1 -1 -1 -1 -1
[6,] 8 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[7,] 9 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

$keyQpow
# Density dependent catchability power parameters (if any).
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[2,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[3,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[4,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[5,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[6,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[7,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

$keyVarF
# Coupling of process variance parameters for log(F)-process (normally only first row is used)
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,] 0 0 0 0 0 0 0 0 0 0 0
[2,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[3,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[4,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[5,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[6,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[7,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

$keyVarLogN
# Coupling of process variance parameters for log(N)-process
0 1 1 1 1 1 1 1 1 1 1

$keyVarObs
# Coupling of the variance parameters for the observations.
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,] 0 1 2 2 2 2 2 2 2 2 2
[2,] 3 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[3,] 4 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[4,] 5 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[5,] 6 7 8 8 8 8 -1 -1 -1 -1 -1
[6,] 9 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[7,] 10 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

$sobsCorStruct
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). |
Possible values are: "ID" "AR" "US"
ID ID ID ID AR ID ID
ID ID ID ID AR ID ID

$keyCorObs
# Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they cannot).
1-2 2-3 3-4 4-5 5-6 6-7 7-8 8-9 9-10 10-11
[1,] NA NA NA NA NA NA NA NA NA NA
[2,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[3,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[4,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[5,] 0 1 1 1 1 -1 -1 -1 -1 -1
[6,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[7,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

$stockRecruitmentModelCode
# Stock recruitment code (0 for plain random walk, 1 for Ricker, and 2 for Beverton-Holt).
0

$noScaledYears
# Number of years where catch scaling is applied.
0

$keyScaledYears

```

```

# A vector of the years where catch scaling is applied.
Numeric(0)
$keyParScaledYA
# A matrix specifying the couplings of scale parameters (nrow = no scaled years, ncol = no ages).
<0 x 0 matrix>
$fbarRange
# lowest and highest age included in Fbar
3 7
$keyBiomassTreat
# To be defined only if a biomass survey is used (0 = SSB index; 1 = catch index; 2 = FSB index; 3 =
total catch; 4 = total landings; 5 = TSB index).
-1 2 2 2 -1 -1 -1
$obsLikelihoodFlag
# Option for observational likelihood | Possible values are: "LN" "ALN"
LN LN LN LN LN LN LN
$fixVarToweight
# If weight attribute is supplied for observations this option sets the treatment (0 relative weight,
1 fix variance to weight).
0
$fracMixF
# The fraction of t(3) distribution used in logF increment distribution
0
$fracMixN
# The fraction of t(3) distribution used in logN increment distribution
0 0 0 0 0 0 0 0 0 0 0
$fracMixObs
# A vector with same length as number of fleets, where each element is the fraction of t(3) distribution
used in the distribution of that fleet
0 0 0 0 0 0 0
$constRecBreaks
# Vector of break years between which recruitment is at constant level. The break year is included in
the left interval. (This option is only used in combination with stock recruitment code =3)
numeric(0)
$predVarObsLink
# Coupling of parameters used in a prediction-variance link for observations
[,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,] -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
[2,] NA NA NA NA NA NA NA NA NA NA NA
[3,] NA NA NA NA NA NA NA NA NA NA NA
[4,] NA NA NA NA NA NA NA NA NA NA NA
[5,] -1 -1 -1 -1 -1 -1 NA NA NA NA NA
[6,] NA NA NA NA NA NA NA NA NA NA NA
[7,] NA NA NA NA NA NA NA NA NA NA NA
$hockeyStickCurve
20
$stockWeightModel
0
$keyStockWeightMean
NA NA NA NA NA NA NA NA NA NA NA
$keyStockWeightObsvar
NA NA NA NA NA NA NA NA NA NA NA
$catchWeightModel
0
$keyCatchWeightMean
NA NA NA NA NA NA NA NA NA NA NA
$keyCatchWeightObsvar
NA NA NA NA NA NA NA NA NA NA NA
$matureModel
0
$keyMatureMean
NA NA NA NA NA NA NA NA NA NA NA
$mortalityModel

```



```
0
$keyMortalityMean
NA NA NA NA NA NA NA NA NA NA NA
$keyMortalityObsVar
NA NA NA NA NA NA NA NA NA NA
$keyXtraSd
[,1] [,2] [,3] [,4]
```

Table 17.14. Sole in Division 7.d. SAM model fitting diagnostics of the 2025 assessment.

Model fitting		
log(Lik)	#par	AIC
-487.9265	26	1027.853

Table 17.15. Sole in Division 7.d. SAM model survey catchability of the 2025 assessment.

estimate	1	2	3	4	5	6	7	8	9	10	11
BE-CBT-LPUE	-4.4123	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-CBT	-2.5819	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-COTB-model	-4.3931	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-BTS-Q3	-7.9948	-7.1636	-7.4916	-7.7378	-7.8837	-7.8837	NA	NA	NA	NA	NA
UK(E&W)-YFS	-9.6638	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-YFS	-11.8197	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

sd	1	2	3	4	5	6	7	8	9	10	11
BE-CBT-LPUE	0.0731	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-CBT	0.0682	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-COTB-model	0.0898	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-BTS-Q3	0.1996	0.0893	0.0963	0.0785	0.0732	0.0732	NA	NA	NA	NA	NA
UK(E&W)-YFS	0.1352	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-YFS	0.1667	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Table 17.16. Sole in Division 7.d. Assessment summary.

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings^	Dis-cards*^	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
1982	9823	14397	21102	11459	13237	15290	3190	195	0.28	0.34	0.41

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings^	Dis-cards*^.^	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
1983	15212	21872	31448	11970	13712	15708	3458	101	0.30	0.36	0.42
1984	15762	22511	32151	12278	13952	15856	3574	141	0.32	0.37	0.44
1985	11191	16067	23066	12962	14661	16583	3838	242	0.31	0.37	0.43
1986	17283	24761	35472	12606	14216	16031	3932	145	0.35	0.40	0.46
1987	10828	15388	21870	12296	13829	15554	4790	197	0.40	0.46	0.54
1988	19162	26470	36566	11494	12984	14667	3853	198	0.39	0.45	0.53
1989	12739	17694	24575	10141	11378	12766	3805	192	0.41	0.47	0.55
1990	32246	44503	61418	10721	12059	13564	3647	341	0.38	0.43	0.50
1991	25923	35406	48356	11278	12672	14238	4351	368	0.37	0.43	0.50
1992	21478	29729	41149	12744	14516	16534	4072	276	0.34	0.40	0.46
1993	9836	13803	19371	14187	16095	18260	4299	273	0.32	0.38	0.44
1994	20322	27895	38290	13237	14922	16822	4383	123	0.35	0.40	0.47
1995	16872	23242	32016	12639	14101	15732	4420	283	0.38	0.44	0.50
1996	13761	18925	26027	12284	13690	15257	4797	174	0.43	0.50	0.57
1997	19660	27732	39118	11351	12623	14037	4764	148	0.46	0.53	0.61
1998	14241	19632	27063	10463	11662	12997	3363	128	0.43	0.50	0.57
1999	25248	34602	47420	9904	11173	12604	4135	247	0.42	0.48	0.55
2000	28309	38638	52736	10767	12058	13504	3475	202	0.37	0.43	0.50
2001	23571	32599	45085	12786	14431	16289	4025	317	0.35	0.40	0.47
2002	36647	50563	69763	14125	15884	17861	4733	445	0.34	0.39	0.45
2003	17677	24154	33004	16725	18754	21029	6977	583	0.37	0.43	0.50
2004	16545	23160	32419	13360	14913	16647	5819	258	0.37	0.42	0.49
2005	35677	48995	67286	13956	15521	17261	4748	344	0.34	0.40	0.46
2006	30754	41984	57313	13606	15148	16866	4831	315	0.38	0.43	0.50
2007	14745	20566	28684	13681	15337	17194	5420	332	0.40	0.46	0.54
2008	18482	25787	35979	13482	15165	17058	4963	183	0.38	0.45	0.52
2009	31648	44895	63689	12953	14694	16669	4828	287	0.37	0.44	0.52
2010	31258	44044	62062	12684	14384	16312	4108	273	0.34	0.41	0.48

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings [^]	Dis-cards ^{*,^}	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
2011	25710	36557	51981	13546	15297	17274	4136	342	0.31	0.36	0.43
2012	13493	19201	27322	14214	16093	18221	4057	444	0.28	0.33	0.39
2013	9338	13310	18970	15711	17799	20164	4295	180	0.26	0.31	0.37
2014	10511	14959	21290	14522	16725	19262	4626	216	0.27	0.31	0.37
2015	11656	16909	24531	12707	14672	16940	3385	250	0.25	0.30	0.36
2016	8078	11546	16502	12265	14206	16455	2434	106	0.22	0.27	0.32
2017	13802	19591	27808	10501	12273	14346	2090	156	0.20	0.25	0.30
2018	12425	17644	25053	10215	11984	14061	2395	263	0.21	0.25	0.30
2019	15001	21409	30556	9483	11167	13149	1648	403	0.189	0.23	0.28
2020	6900	10026	14568	9325	11114	13246	1563	409	0.188	0.23	0.28
2021	8438	12215	17682	10067	12095	14531	1567	347	0.182	0.23	0.28
2022	5753	8914	13812	8056	9817	11962	1700	313	0.188	0.23	0.29
2023	8072	14149	24799	8244	10292	12849	1292	207	0.171	0.22	0.27
2024	5806	16044	44336	8204	10370	13107	1112	462	0.160	0.21	0.27
2025		14696**		8127	10542	13704					

* Discard estimates prior to 2004 assume the average discard proportion by age for 2004–2008 (WKNSEA; ICES, 2021).

** Median recruitment resampled (weighted by the inverse of the sd of the recruitment estimates) from the years 2012–2024 (geometric mean shown).

[^] Landings and discards for ages 1–11+ only, as used in the assessment.

Table 17.17. Sole in Division 7.d. Stock numbers for the 2024 SAM assessment (in thousands).

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	14397	21872	22511	16067	24761	15388	26470	17694	44503	35406	29729	13803	27895
2	17883	12476	20039	20842	13815	23480	13107	24850	15043	41754	30338	28665	11616
3	20877	13230	9772	15849	15425	10204	18053	8796	18119	10423	30439	22327	22186
4	5052	13256	7984	5763	9278	8126	5176	9278	4037	10011	5477	17938	13688
5	3217	2990	8309	4729	3557	5430	4200	2864	4702	2072	5519	3188	10759
6	3058	2372	1900	5909	3230	2404	3171	2411	1487	2650	1182	3115	1985
7	1875	2122	1564	1048	3894	2178	1383	1924	1476	978	1572	783	2133
8	921	1207	1417	1011	704	2498	1222	844	1176	945	647	988	552

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
9	583	621	782	1030	790	502	1595	821	572	802	619	469	678
10	408	414	444	535	799	578	351	1106	556	385	503	400	341
11+	988	1000	1028	1029	1266	1504	1388	1223	1648	1440	1148	1061	1089

Age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	23242	18925	27732	19632	34602	38638	32599	50563	24154	23160	48995	41984	20566
2	25609	19690	17226	26889	17125	31339	34860	29436	46895	20759	19233	44015	38509
3	9737	19885	14782	13788	23102	12285	22778	24072	19700	29916	14150	12334	30302
4	13408	5675	10154	7040	7224	12536	6025	12894	13109	9683	16029	8389	6816
5	7852	7728	3019	4763	3470	3889	7264	3627	7740	8191	5503	9803	4919
6	6390	4630	4238	1525	2479	1895	2368	4248	2232	4648	4395	3466	5869
7	1453	3985	2799	2463	934	1365	1191	1551	3179	1535	2877	2958	2237
8	1414	1058	2336	1664	1572	546	831	784	1107	1814	1120	1920	1774
9	410	1024	801	1398	1075	943	345	604	591	827	1159	793	1245
10	473	307	677	573	848	661	557	251	464	490	529	791	532
11+	1019	1070	914	1054	1075	1213	1260	1245	1103	1120	1026	1088	1263

Age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	25787	44895	44044	36557	19201	13310	14959	16909	11546	19591	17644	21409	10026
2	17907	23046	38825	40132	33435	16455	11814	13404	15491	9863	18205	15423	19623
3	27576	13588	16074	27118	31421	27577	13500	9405	10286	13329	7659	14087	11729
4	17588	17352	8192	9399	16805	21547	21131	9067	6460	6957	10745	5100	10041
5	3494	9609	9701	4594	5447	10224	14149	13237	5994	4487	4836	7428	3640
6	2919	1968	5325	5631	2943	3526	6958	8914	8052	3969	3228	3521	5151
7	3086	1685	1107	2821	3337	2068	2508	4614	5693	5296	2695	2133	2702
8	1281	1823	910	671	1706	2250	1450	1722	3003	3909	3813	1770	1524
9	1003	810	1090	533	433	1074	1560	982	1175	2084	2856	2694	1286
10	744	684	480	675	317	296	730	1074	742	845	1558	2107	1951
11+	1130	1290	1286	1112	1233	1052	1032	1248	1566	1515	1726	2286	3223

Age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Age	2021			2022			2023			2024			
1	12215			8914			14149			16044			
2	8485			11001			7665			12603			
3	15534			6277			8071			5515			
4	7963			11377			4138			5370			
5	6912			5335			8182			2789			
6	2620			4821			3346			5823			
7	3652			1950			3444			2447			
8	2100			2863			1326			2644			
9	1181			1721			2235			1002			
10	994			961			1379			1842			
11+	3658			3485			3276			3680			

Table 17.18. Sole in Division 7.d. Fishing pressure (F) at age for the 2024 SAM assessment.

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	0.010	0.009	0.009	0.009	0.009	0.009	0.010	0.010	0.011	0.011	0.011	0.010	0.010
2	0.175	0.165	0.173	0.191	0.193	0.208	0.226	0.229	0.235	0.221	0.202	0.185	0.160
3	0.377	0.399	0.436	0.469	0.516	0.552	0.572	0.573	0.523	0.510	0.446	0.429	0.443
4	0.414	0.403	0.413	0.417	0.454	0.500	0.498	0.537	0.504	0.498	0.472	0.449	0.485
5	0.257	0.291	0.289	0.302	0.337	0.402	0.431	0.489	0.462	0.445	0.446	0.430	0.435
6	0.319	0.344	0.366	0.321	0.348	0.430	0.387	0.378	0.337	0.349	0.316	0.289	0.323
7	0.319	0.344	0.366	0.321	0.348	0.430	0.387	0.378	0.337	0.349	0.316	0.289	0.323
8	0.262	0.253	0.233	0.215	0.247	0.271	0.254	0.267	0.282	0.311	0.291	0.271	0.261
9	0.262	0.253	0.233	0.215	0.247	0.271	0.254	0.267	0.282	0.311	0.291	0.271	0.261
10	0.234	0.235	0.235	0.209	0.267	0.292	0.272	0.272	0.290	0.285	0.252	0.226	0.272
11+	0.234	0.235	0.235	0.209	0.267	0.292	0.272	0.272	0.290	0.285	0.252	0.226	0.272

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	0.010	0.010	0.010	0.010	0.010	0.010	0.011	0.011	0.011	0.012	0.012	0.013	0.014
2	0.159	0.155	0.153	0.158	0.196	0.217	0.250	0.282	0.293	0.283	0.285	0.251	0.235

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
3	0.514	0.563	0.595	0.577	0.548	0.551	0.500	0.518	0.561	0.506	0.473	0.454	0.415
4	0.505	0.575	0.609	0.577	0.537	0.470	0.421	0.407	0.413	0.440	0.427	0.462	0.509
5	0.476	0.528	0.593	0.532	0.491	0.436	0.440	0.446	0.435	0.472	0.408	0.425	0.445
6	0.349	0.407	0.422	0.401	0.409	0.351	0.324	0.289	0.376	0.353	0.339	0.413	0.477
7	0.349	0.407	0.422	0.401	0.409	0.351	0.324	0.289	0.376	0.353	0.339	0.413	0.477
8	0.265	0.304	0.340	0.349	0.376	0.329	0.237	0.214	0.223	0.293	0.314	0.351	0.395
9	0.265	0.304	0.340	0.349	0.376	0.329	0.237	0.214	0.223	0.293	0.314	0.351	0.395
10	0.294	0.342	0.340	0.351	0.321	0.276	0.249	0.260	0.317	0.388	0.342	0.362	0.373
11+	0.294	0.342	0.340	0.351	0.321	0.276	0.249	0.260	0.317	0.388	0.342	0.362	0.373

age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	0.015	0.015	0.016	0.016	0.017	0.017	0.018	0.019	0.019	0.020	0.021	0.022	0.022
2	0.218	0.210	0.193	0.165	0.141	0.127	0.129	0.134	0.129	0.141	0.158	0.165	0.171
3	0.394	0.389	0.367	0.332	0.300	0.271	0.278	0.267	0.256	0.242	0.260	0.260	0.267
4	0.480	0.481	0.461	0.425	0.392	0.359	0.350	0.313	0.277	0.268	0.269	0.265	0.263
5	0.464	0.424	0.414	0.393	0.356	0.335	0.340	0.319	0.281	0.253	0.248	0.243	0.244
6	0.444	0.456	0.392	0.335	0.305	0.298	0.303	0.301	0.259	0.241	0.235	0.195	0.189
7	0.444	0.456	0.392	0.335	0.305	0.298	0.303	0.301	0.259	0.241	0.235	0.195	0.189
8	0.348	0.378	0.357	0.340	0.308	0.283	0.270	0.235	0.214	0.195	0.189	0.160	0.132
9	0.348	0.378	0.357	0.340	0.308	0.283	0.270	0.235	0.214	0.195	0.189	0.160	0.132
10	0.329	0.331	0.286	0.241	0.224	0.216	0.235	0.243	0.239	0.194	0.175	0.137	0.108
11+	0.329	0.331	0.286	0.241	0.224	0.216	0.235	0.243	0.239	0.194	0.175	0.137	0.108

age	2021	2022	2023	2024
1	0.023	0.024	0.025	0.026
2	0.182	0.200	0.212	0.225
3	0.269	0.284	0.297	0.305
4	0.260	0.256	0.250	0.250
5	0.240	0.251	0.222	0.210
6	0.179	0.187	0.155	0.135

age	2021	2022	2023	2024
7	0.179	0.187	0.155	0.135
8	0.121	0.114	0.091	0.077
9	0.121	0.114	0.091	0.077
10	0.083	0.070	0.054	0.046
11+	0.083	0.070	0.054	0.046

Table 17.19. Sole in Division 7.d. Spawning stock biomass (SSB; tonnes) at age for the 2024 SAM assessment.

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1583	1117	1859	1977	1296	2140	1035	1989	1395	3474	2412	2203	905
3	4667	3019	2274	3616	3250	2394	4053	1683	4284	2157	6133	4375	4103
4	1707	4327	2713	1759	3073	2590	1560	2726	1267	2893	1588	4805	3482
5	1323	1288	3361	1853	1377	2044	1609	895	1747	782	1702	1138	3298
6	1444	1141	914	2570	1379	1079	1430	953	557	1153	526	1321	778
7	915	1023	723	600	1888	993	704	828	769	458	682	467	900
8	406	548	660	476	317	1044	544	345	474	353	259	473	242
9	284	323	401	592	384	226	855	410	270	337	275	234	318
10	196	251	260	331	434	323	206	628	299	191	242	319	175
11+	711	676	787	886	818	996	989	923	996	874	696	761	721

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	2348	1691	1369	2223	1343	2309	2753	2481	3977	1562	1570	3313	2857
3	1729	3512	2761	2359	4336	2159	4778	4872	4459	4927	2460	2065	4684
4	3347	1406	2564	1791	1678	3033	1851	3540	4103	2092	4032	1941	1440
5	2133	2211	937	1451	959	1377	2227	1221	3018	2105	1927	3005	1346
6	2345	1653	1602	578	769	788	871	1763	1004	1325	1947	1220	1849
7	575	1594	1209	1047	301	643	486	648	582	628	1341	1177	774
8	472	371	818	601	464	202	269	267	455	925	576	893	786
9	195	431	390	656	373	351	164	274	243	323	693	455	617

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
10	243	148	334	259	330	315	296	109	243	252	372	392	211
11+	714	672	639	697	621	882	737	708	671	775	603	687	774

age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1338	1808	2654	2531	1843	907	582	725	829	706	1071	850	1009
3	4820	2450	2854	4690	5117	4136	1789	1531	1997	2158	1247	2294	1532
4	4153	4398	2045	2256	3678	5047	4767	2124	1693	1529	2548	1136	1957
5	844	2684	2748	1395	1453	3362	3884	3801	1709	1162	1389	1960	929
6	987	677	1933	1903	974	1199	2407	2745	2665	1210	1046	1074	1406
7	1028	512	402	1047	1291	908	993	1721	2089	1711	924	655	856
8	557	798	338	323	655	936	622	578	1345	1501	1266	623	562
9	374	285	459	218	202	463	690	391	631	996	1060	770	590
10	530	299	260	309	174	123	432	408	338	429	634	760	675
11+	533	782	691	624	706	719	558	648	909	870	800	1044	1597

age	2021	2022	2023	2024
1	0	0	0	0
2	454	531	431	835
3	2472	745	1233	847
4	1682	2119	894	1155
5	1837	1206	1881	695
6	815	1292	1007	1782
7	1209	531	1133	915
8	779	876	518	973
9	470	597	854	503
10	413	338	516	866
11+	1963	1583	1826	1798